

OMAR NOUH

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SUMMARY

Third-year Computer and Artificial Intelligence Engineering student focused on cross-stack development, from hardware architecture to scalable software. Experienced in C/C++, Python, and C# through the hands-on engineering of custom OS kernels, normalized enterprise databases, and deep learning pipelines.

EDUCATION

Ain Shams University | Cairo, EG

B.Sc. Computer & Artificial Intelligence Engineering, (Dual Degree — University of East London) | Expected 2028

• **GPA:** 3.6

• **Coursework:** Operating Systems, Compiler Design, Logic Design, Data Structures & Algorithms, Artificial Intelligence, Discrete Mathematics

Collège de la Sainte Famille

Baccalauréat Français — Highest Honors (Summa Cum Laude / Félicitations du Jury) | 2024

TECHNICAL SKILLS

Languages

C / C++, Python, Java, C#, VHDL, MIPS Assembly, JavaScript, SQL

Systems & Architecture

Memory Management, FOS Kernel, Compiler Design, CPU Scheduling, RTL Design, FPGA, Digital Logic, Hardware-Software Interface

Tools & Frameworks

Git/GitHub, Vivado, Logisim, LTSpice, .NET, WinForms, PyTorch, scikit-learn, Microsoft Excel, Office 365

PROFESSIONAL EXPERIENCE

Dell Technologies | Artificial Intelligence Engineer Intern | Aug 2026

- Analyzed enterprise cloud database architectures, evaluating hardware-software integration constraints across Dell infrastructure platforms.
- Developed a foundational understanding of high-availability cloud deployments, bridging physical hardware limitations with scalable software solutions.

e& | Enterprise Security Intern | Aug 2025

- Evaluated enterprise-scale telecommunications and network infrastructure, analyzing system-level vulnerabilities and cyber defense architectures.
- Assessed Governance, Risk, and Compliance (GRC) frameworks, bridging the gap between low-level information security systems and high-level enterprise protocols.

Future Career | Business Process Engineering Intern | Jul 2024

- Architected API-driven automated workflows utilizing Activepieces, integrating third-party services to streamline complex business logic and data routing.
- Engineered programmatic data-parsing pipelines that successfully reduced manual data entry overhead by 50% and accelerated client engagement metrics by 25% within a two-month lifecycle.

SELECTED ENGINEERING PROJECTS

Operations & Inventory Tracker | Microsoft Excel, Javascript

- Designed and built a multi-sheet relational tracking system in Excel to monitor raw materials, production stages, and order fulfillment for a clothing factory.
- Programmed dynamic data validation, advanced lookups (INDEX/MATCH), and conditional formatting to eliminate manual entry errors and ensure data integrity.
- Structured automated Pivot Tables and summary dashboards to give management clear, real-time visibility into operational bottlenecks and inventory levels.

Operating System Kernel (FOS) | C, Virtual Memory, CPU Scheduling

- Engineered a custom Multi-Level Feedback Queue (MLFQ) CPU scheduler in C, implementing dynamic priority boosting and aging to optimize execution and eliminate process starvation across queues.
- Architected a dynamic kernel heap allocator, managing direct virtual-to-physical address translation, frame allocation, and hierarchical page-directory and page-table structures.
- Developed the core memory management subsystem by implementing an Nth-Chance Clock page replacement algorithm, tightly integrated with working-set tracking, page-file I/O, and robust page-fault handlers.

Custom Tiny Language Compiler | C/C++, Lexical & Syntax Analysis, Automata Theory

- Built a complete compiler frontend from scratch in C/C++, engineering a hand-coded lexical analyzer based on custom Regular Expressions and Deterministic Finite Automata (DFAs) to ensure robust tokenization and error recovery.
- Designed a formal Context-Free Grammar (CFG) to parse and validate complex programmatic constructs, including functions, control flow loops, conditionals, and multi-variable arithmetic expressions.
- Implemented a recursive-descent parser to execute strict syntax analysis, dynamically generating visual parse trees to verify code structure and structural correctness.

Enterprise Cinema Booking System | C#, .NET, SQL Server, WinForms

- Architected a highly normalized (3NF) relational database schema in SQL Server encompassing 12 tables, enforcing strict primary/foreign key constraints and role-based access control.
- Engineered a strict, zero ad-hoc SQL data access layer utilizing modular stored procedures and scalar functions to securely bridge the database and the application layer.
- Developed an end-to-end layered desktop application in C# .NET to manage the complete transaction lifecycle, handling user authentication, secure payment flows, and concurrent order state management.

32-Bit Single-Cycle MIPS Processor | VHDL, Vivado, Computer Architecture

- Designed and simulated a modular 32-bit CPU datapath utilizing VHDL and Xilinx toolchains, integrating a custom ALU, dynamic register files, and distinct instruction/data memory blocks.
- Implemented a hardwired control unit capable of decoding R, I, and J instruction formats, driving opcode-dependent execution signals and data flow across the processor.
- Engineered comprehensive simulation testbenches to validate opcode branching logic, analyze low-level datapath timing constraints, and debug ALU flags and structural register hazards.

Machine Learning Scene Classification | Python, scikit-learn, PyTorch

- Built an end-to-end machine learning pipeline for image scene classification, engineering robust data preprocessing workflows to handle image distortions and categorical encoding, optimizing classification models by implementing advanced feature engineering.
- Executed comprehensive fine-tuning on complex deep learning architectures (including Convolution and Transformers) and model evaluation techniques on custom structured datasets, successfully achieving 60.3% validation accuracy.

LEADERSHIP & ACTIVITIES: Active in Competitive Programming, Model United Nations (MUN), and Triathlons.

LANGUAGES: English | Native — French | Native — Arabic | Native — Spanish | Proficient